A natural proof of Quadratic Reciprocity

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1 Introduction

The purpose of this text is to give a proof of Quadratic Reciprocity which is, in some sense, an *additive version of the standard proof of Gauss's Lemma*.[1] Here is a more precise formulation:

A common proof of Quadratic Reciprocity proceeds in three steps:

1. Prove Gauss's Lemma.

2. Derive Corollary 5 below from Gauss's Lemma.

3. Derive Quadratic Reciprocity from Corollary 5.

This is done in Hardy and Wright [1], in Niven, Zuckerman and Montgomery [2], and in Wallach [3] (and probably in many other texts). Steps 1 and 3 are fairly standard, but there are numerous ways of doing Step 2. Our main observation is that a minor variation of the standard argument used in Step 1 works for Step 2.

To elaborate: the assumptions of Gauss's Lemma and of Theorem 2 are that p is an odd prime and a an integer prime to p. The standard argument in question goes as follows: we set

$$I = \{1, 2, \dots, \tfrac{p-1}{2}\}, \quad U = \{a, 2a, 3a, \dots, \tfrac{p-1}{2}\, a\}, \tag{1}$$

we define two other sets M and V,[2] we prove $M = I$ and $V = U$, and we derive the conclusion from the equalities

$$\prod_{i\in I} i = \prod_{m\in M} m \quad \text{and} \quad \prod_{u\in U} u = \prod_{v\in V} v. \tag{2}$$

Our point is that Step 2 can be easily accomplished via the additive analog

$$\sum_{i\in I} i = \sum_{m\in M} m \quad \text{and} \quad \sum_{u\in U} u = \sum_{v\in V} v \tag{3}$$

of the equalities (2).

Step 3 is basically the same in [1], [2] and [3], and also in this text. Corollary 5 is a corollary to Theorem 2. Theorem 2 is implicit in the proof of Theorem 3.3 in [2]; it is even more implicit in [1] (Section 6.13); it appears as Lemma 10 in [3]. Theorem 2 yields the most transparent proof I know of the second supplement[3] to the law of Quadratic Reciprocity (this argument was already given in [2]—see again the proof of Theorem 3.3 there).

[1] This text is available at `https://github.com/Pierre-Yves-Gaillard/quadratic-reciprocity`

[2] The sets M and V are respectively defined in (5) and (7) below.

[3] This is the equality $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$.

In the last two sections we give two proofs of Gauss's and a generalization of Euler's Criterion.

Until further notice, p and q are distinct odd primes, and a is an integer prime to p. Recall that the **Legendre symbol** $(\frac{a}{p})$ is equal to 1 if a is a square $(\bmod p)$, and to -1 if a is a non-square $(\bmod p)$. (Since the integers x and y have the same square $(\bmod p)$ if and only if $y \equiv \pm x \pmod{p}$, we see that there are $(p-1)/2$ nonzero square residues, and $(p-1)/2$ non-square residues $(\bmod p)$.) Our main goal is to prove Quadratic Reciprocity:

Theorem 1 (Quadratic Reciprocity)**.** *If p and q are distinct odd primes, then we have $(\frac{p}{q})(\frac{q}{p}) = (-1)^{(p-1)(q-1)/4}$.*

2 Proof of Gauss's Lemma and Theorem 2

Let $\lfloor x \rfloor$ denote the floor of the real number x (the greatest integer less than or equal to x), and set

$$f(a,p) = \sum_{i=1}^{(p-1)/2} \left\lfloor \frac{ia}{p} \right\rfloor.$$

Quadratic Reciprocity will follow from the theorem below:

Theorem 2. *If p is an odd prime and a an integer prime to p, then we have $(\frac{a}{p}) = (-1)^{\ell}$ with*

$$\ell = (a+1)\,\frac{p^2-1}{8} + f(a,p).$$

Remark 3. *We have*

$$\frac{p^2-1}{8} = \sum_{i=1}^{(p-1)/2} i;$$

in particular $(p^2-1)/8$ is an integer. (One can also observe that $\frac{p-1}{2}\frac{p+1}{2}$ is even.)

The proof will proceed by comparing multiplicative and additive analogs of the same combinatorial identities.

Corollary 4. *We have $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$.*

This follows from the fact that $f(2,p) = 0$, because, for $1 \le i \le (p-1)/2$, we have $2i \le p-1 < p$, so $\lfloor 2i/p \rfloor = 0$.

Corollary 5. *We have $(\frac{a}{p}) = (-1)^{f(a,p)}$ if a is odd.*

To prove Theorem 2 we shall use Euler's Criterion:

Theorem 6 (Euler's Criterion)**.** *Recall that p is an odd prime and a is an integer prime to p. Set $n = (p-1)/2$. Then we have $(\frac{a}{p}) \equiv a^n \pmod{p}$.*

A generalization of Euler's Criterion is given in the last section.

Corollary 7. *We have $(\frac{ab}{p}) = (\frac{a}{p})(\frac{b}{p})$ if a and b are integers prime to p. We also have $(\frac{1}{p}) = 1$.*

To prove Euler's Criterion we use the following theorem:

Theorem 8. *The multiplicative group $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$ of the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is cyclic (of order $p-1$).*

For a proof of Theorem 8, we refer the reader to Weil's book [4], Theorem X.1. A generalization of Euler's Criterion is given in the last section.

Proof of Euler's Criterion (Theorem 6). If b is in $\mathbb{F}_p^*$, we have $b^n = \pm 1$ (because $(b^n)^2 = b^{2n} = 1$) and we must show that $b^n = 1$ if and only if b is a square. Let ζ be a generator of $\mathbb{F}_p^*$. Then $b = \zeta^j$ for some integer j. Then we get $b^n = \zeta^{jn}$, and $b^n = 1$ if and only if $2n$ divides jn, which means j is even; this, in turn, means b is a square. $\square$

For $i \in I$ (see (1)), let r_i be the remainder when ia is divided by p, so that we have

$$ia = p\left\lfloor \frac{ia}{p} \right\rfloor + r_i. \tag{4}$$

In particular, the r_i are distinct and satisfy $1 \leq r_i \leq p-1$. Set $J = \{i \in I \mid r_i \leq n\}$, $K = \{i \in I \mid r_i \geq n+1\}$. We denote the cardinality of the set X by $|X|$ and we indicate by $X = Y \sqcup Z$ the fact that X is the disjoint union of Y and Z.

Theorem 9 (Gauss's Lemma)**.** *We have $(\frac{a}{p}) = (-1)^{|K|}$.*

Proof of Gauss's Lemma (Theorem 9) and of Theorem 2. The equality $I = J \sqcup K$ is clear. Set

$$M_1 = \{r_j \mid j \in J\}, \quad M_2 = \{p - r_k \mid k \in K\}, \quad M = M_1 \cup M_2. \tag{5}$$

We claim

$$I = M_1 \sqcup M_2 = M. \tag{6}$$

To prove (6), note that, since the r_i are distinct (indeed if $r_i = r_j$ then $p \mid a(i-j)$, and since $\gcd(a,p) = 1$ and $|i-j| < p$ we get $i = j$), it suffices to show $r_j \neq p - r_k$ for $j \in J, k \in K$. But $r_j = p - r_k$ would imply $0 \equiv r_j + r_k \equiv a(j+k) \pmod p$, hence $j + k \equiv 0 \pmod p$, hence p divides $j+k$, contradicting the inequalities $2 \leq j+k \leq p-1$. This proves (6).

The sets I, U, M and V in (2) and (3) are defined by (1), (5) and

$$V = \left\{ p\left\lfloor \frac{ia}{p} \right\rfloor + r_i \;\middle|\; i \in I \right\}. \tag{7}$$

The equalities $M = I$ and $V = U$ follow respectively from (6) and (4).

In the remainder of this section i, j and k run respectively over I, J and K.

In view of (1), (5) and (7), the first and second equalities in (2) give respectively

$$n! = \prod i = \left(\prod r_j\right)\left(\prod (p - r_k)\right) \equiv (-1)^{|K|} \prod r_i \pmod p \tag{8}$$

and

$$n!\, a^n = \prod ia = \prod \left(p\left\lfloor \frac{ia}{p} \right\rfloor + r_i\right) \equiv \prod r_i \pmod p. \tag{9}$$

Similarly, the first and second equalities in (3) give respectively

$$\frac{p^2-1}{8} = \sum i = \sum r_j + \sum (p - r_k) = \sum r_j + p\,|K| - \sum r_k \equiv \sum r_i + |K| \pmod 2 \tag{10}$$

and

$$a\,\frac{p^2-1}{8}=\sum_{i=1}^{n} ia = p\,f(a,p)+\sum r_i \equiv f(a,p)+\sum r_i \pmod 2. \tag{11}$$

Now Gauss's Lemma (Theorem 9) follows from Euler's Criterion (Theorem 6), (8) and (9). To prove Theorem 2, it suffices to show $|K| \equiv \ell \pmod 2$ (in the notation of Theorems 9 and 2). We have

$$|K| \equiv \frac{p^2-1}{8}+\sum r_i \equiv \frac{p^2-1}{8}+a\,\frac{p^2-1}{8}+f(a,p)\equiv \ell \pmod 2,$$

the first congruence following from (10) and the second from (11). □

3 Proof of Quadratic Reciprocity

We prove Quadratic Reciprocity (Theorem 1). Recall that p and q are distinct odd primes. By Corollary 5, it suffices to show

$$f(p,q)+f(q,p)=\frac{(p-1)(q-1)}{4}\ .$$

For any subset S of $\mathbb{R}^2$, put $S^* = S \cap \mathbb{Z}^2$. Consider the open rectangle

$$R=\{(x,y)\in\mathbb{R}^2 \mid 0<x<\tfrac{p}{2}, 0<y<\tfrac{q}{2}\},$$

the diagonal

$$D=\{(x,y)\in R \mid py=qx\},$$

the semi-open triangle above the diagonal

$$A=\{(x,y)\in R \mid py\ge qx\},$$

and the semi-open triangle below the diagonal

$$B=\{(x,y)\in R \mid py\le qx\}.$$

Recall that, if X, Y, Z are sets, we denote the cardinality of X by $|X|$ and we indicate by $X = Y \sqcup Z$ the fact that X is the disjoint union of Y and Z. We have $D^* = \varnothing$. Indeed, $py = qx$ with $(x,y)\in R^*$ would imply that p divides x (because p and q are relatively prime), which is impossible since $0 < x < p/2$. We get

$$A^*\cap B^* = D^* = \varnothing, \quad R^* = A^* \sqcup B^*, \quad |A^*|+|B^*| = |R^*| = \frac{(p-1)(q-1)}{4}\ .$$

From the equations displayed above, it is enough to show $f(p,q) = |A^*|$ and $f(q,p) = |B^*|$. Let us prove $f(q,p) = |B^*|$. Recall the definition of Iverson's bracket: if (S) is a statement, then $[(S)] = 1$ if (S) is true, and $[(S)] = 0$ if (S) is false. Recall also that, if $t \ge 0$, then $\lfloor t\rfloor$ is the number of positive integers $\le t$:

$$\lfloor t\rfloor = \sum_{i\in\mathbb{Z}}\,[1\le i\le t] \quad \text{for } t\ge 0.$$

We have

$$|B^*| = \sum_{i=1}^{(p-1)/2}\sum_{j\in\mathbb{Z}}\left[1\le j\le \frac{iq}{p}\right] = \sum_{i=1}^{(p-1)/2}\left\lfloor\frac{iq}{p}\right\rfloor = f(q,p).$$

The equality $f(p,q) = |A^*|$ is proved similarly. This proves Quadratic Reciprocity.

4 Two proofs of Gauss's Lemma

We add two proofs of Gauss's Lemma (or, more precisely, of a version of Gauss's Lemma which is slightly stronger than the one in Theorem 9).

Recall that p is an odd prime, that $n = (p-1)/2$, that $\mathbb{F}_p$ is the field $\mathbb{Z}/p\mathbb{Z}$, and that $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$ is its multiplicative group (see Theorem 8). Let $a \in \mathbb{F}_p^*$ and let $x_1, \ldots, x_n \in \mathbb{F}_p^*$ satisfy

$$\mathbb{F}_p^* = \{x_1, -x_1, \ldots, x_n, -x_n\}.$$

There is a unique permutation σ of $I = \{1, \ldots, n\}$ such that $ax_i = \pm x_{\sigma(i)}$ for all $i \in I$. Recall that $(\frac{a}{p})$ is the Legendre symbol of a and p. Indeed, multiplication by a is a bijection on $\mathbb{F}_p^*$ which permutes the subsets $\{x_i, -x_i\}$. (We defined $(\frac{b}{p})$ for $b \in \mathbb{Z}$ prime to p, but it is clear that, once p is chosen, $(\frac{b}{p})$ depends only on the congruence class $\bar{b}$ of $b \pmod p$, so that $(\frac{\bar{b}}{p})$ is well-defined. Also we regard $(\frac{c}{p})$ as equal to ± 1 sometimes in $\mathbb{Z}$, sometimes in $\mathbb{F}_p$.)

Theorem 10 (Gauss's Lemma, strong form)**.** *In the above setting we have*

$$\left(\frac{a}{p}\right) = \prod_{i=1}^{n} \frac{ax_i}{x_{\sigma(i)}} \ . \tag{12}$$

First proof of Theorem 10. Since the right-hand side is equal to a^n, the result follows from Euler's Criterion (Theorem 6). $\square$

The above proof is especially short, but slightly mysterious. The following proof will be a little bit longer, but hopefully less mysterious. More precisely, we hope the next proof will make clearer where the expression in the right-hand side of (12) comes from.

Lemma 11. *The Legendre symbol $(\frac{a}{p})$ is the sign of the permutation $x \mapsto ax$ of $\mathbb{F}_p^*$.*

Proof. Let ζ be a generator of $\mathbb{F}_p^*$ (see Theorem 8). It suffices to show that the permutation $x \mapsto \zeta x$ of $\mathbb{F}_p^*$ is odd (see Corollary 7) (indeed, both $x \mapsto (\frac{x}{p})$ and the sign of the permutation $y \mapsto xy$ of $\mathbb{F}_p^*$ are multiplicative homomorphisms $\mathbb{F}_p^* \to \{\pm 1\}$, so it suffices to check they agree on a generator). But this permutation is the $2n$-cycle $(\zeta^1, \zeta^2, \ldots, \zeta^{2n})$. $\square$

Theorem 12. *Let σ be a permutation of $\mathbb{F}_p^*$ satisfying $\sigma(-x) = -\sigma(x)$ for all $x \in \mathbb{F}_p^*$. Then there is a unique permutation σ' of $I = \{1, \ldots, n\}$ such that $\sigma(x_i) = \pm x_{\sigma'(i)}$ for all $i \in I$, and we have*

$$\operatorname{sgn}(\sigma) = \prod_{i=1}^{n} \frac{\sigma(x_i)}{x_{\sigma'(i)}} \ .$$

Proof. The existence of σ' is clear. Let K be the set of those $i \in I$ such that $\sigma(x_i) = -x_{\sigma'(i)}$. We must show that the sign of σ is $(-1)^{|K|}$, where $|K|$ is the cardinality of K. It suffices to define permutations $\sigma_1, \ldots, \sigma_m$ of $\mathbb{F}_p^*$ such that:

$$\sigma = \sigma_m \circ \cdots \circ \sigma_1, \quad m = |K| + 2, \tag{13}$$

$$\operatorname{sgn}(\sigma_1) = \operatorname{sgn}(\sigma') = \operatorname{sgn}(\sigma_2), \quad \operatorname{sgn}(\sigma_3) = \operatorname{sgn}(\sigma_4) = \cdots = \operatorname{sgn}(\sigma_m) = -1. \tag{14}$$

To do that, we set

$$\sigma_1(x_i) = x_{\sigma'(i)}, \quad \sigma_1(-x_i) = -x_i, \quad \sigma_2(x_i) = x_i, \quad \sigma_2(-x_i) = -x_{\sigma'(i)},$$

and we define $\sigma_3, \sigma_4, \dots, \sigma_m$ as follows:

σ_3 is the transposition $(x_{\sigma'(k)}, -x_{\sigma'(k)})$ where k is the least element of K,

σ_4 is the transposition $(x_{\sigma'(k)}, -x_{\sigma'(k)})$ where k is the next element of K,

$\dots$,

σ_m is the transposition $(x_{\sigma'(k)}, -x_{\sigma'(k)})$ where k is the largest element of K.

In particular $\sigma_2 \circ \sigma_1$ sends x_i to $x_{\sigma'(i)}$ and $-x_i$ to $-x_{\sigma'(i)}$. We have $\operatorname{sgn}(\sigma_1) \cdot \operatorname{sgn}(\sigma_2) = \operatorname{sgn}(\sigma')^2 = 1$, and each of $\sigma_3, \dots, \sigma_m$ is a transposition, giving $\operatorname{sgn}(\sigma) = (-1)^{|K|}$, and we easily check the equalities (13) and (14). $\square$

Second proof of Theorem 10. Theorem 10 follows from Lemma 11 and Theorem 12. $\square$

5 A generalization of Euler's Criterion

Let m be a positive integer and G the group $\mathbb{Z}/m\mathbb{Z}$. For each integer k we set $kG = \{k\gamma \mid \gamma \in G\}$ and we define $f_k : G \to G$ by $f_k(\gamma) = k\gamma$. Note that kG is a subgroup of G and f_k is an endomorphism of G.

Proposition 13. *If k is an integer and if $d = \gcd(k, m)$, then $\operatorname{Im} f_k = kG = dG = \operatorname{Ker} f_{m/d}$, and the order of this subgroup is $\frac{m}{d}$.*

We get Euler's Criterion by setting $m = p - 1$ and $k = 2$, which gives $d = 2$ and $\frac{m}{d} = (p-1)/2$.

Proof. To prove $kG = dG$, note that we have $k\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$ by definition of d, and that these subgroups of $\mathbb{Z}$ project respectively to kG and to dG in G. To show $dG \subset \operatorname{Ker} f_{m/d}$ note that $\frac{m}{d} d\gamma = m\gamma = 0$ for $\gamma \in G$. Let us check $\operatorname{Ker} f_{m/d} \subset dG$. Let x be an integer such that $\frac{m}{d} x \equiv 0 \pmod m$. We must show that d divides x. But we have $\frac{m}{d} x \in m\mathbb{Z}$, and thus $\frac{x}{d} \in \mathbb{Z}$. It only remains to prove $|dG| = \frac{m}{d}$. Setting $\bar{x} = x \bmod m$ for $x \in \mathbb{Z}$, we get, since $\frac{m}{d}\bar{d} = \bar{0}$,

$$dG = \{\dots, -2\bar{d}, -\bar{d}, \bar{0}, \bar{d}, 2\bar{d}, \dots\} = \{\bar{0}, \bar{d}, 2\bar{d}, \dots, (\tfrac{m}{d} - 1)\,\bar{d}\},$$

and the elements $\bar{0}, \bar{d}, 2\bar{d}, \dots, (\frac{m}{d} - 1)\,\bar{d}$ are distinct because $0 < d < 2d < \cdots < (\frac{m}{d} - 1)d < m$. $\square$

References

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[4] André Weil. *Number Theory for Beginners*. Springer, 1979.